

PathAI

PathAI Student Engagement & Risk Prediction System

This project presents a data-driven system designed to help universities identify disengaged students early and recommend suitable academic pathways.

We developed three core components:

- A dynamic engagement scoring system that tracks student activity over time
- A predictive model that identifies at-risk students before Week 6
- A recommendation engine that suggests suitable courses for future enrollment

The system enables proactive intervention by academic staff, helping reduce dropout rates and improve student success outcomes.

Key insight: declining student activity and low engagement are strong early indicators of withdrawal risk.

Task 1: Engagement Score

We developed a dynamic engagement score (0–100) based on student activity and academic performance.

Features used:

- Weekly click activity (VLE interactions)
- Assessment scores

The engagement score was computed using a weighted combination of normalized activity and performance metrics. This score updates weekly, allowing us to track changes in student behavior over time.

Key observation:

Students with consistently high activity and performance maintain high engagement scores, while disengaged students show a sharp decline early in the course.

Task 2: Disengagement Prediction

We built a binary classification model to predict whether a student will withdraw or fail before Week 6.

Model:

- Logistic Regression with class balancing

Features:

- Total clicks (engagement)
- Average score (performance)

We prioritized recall to ensure that at-risk students are not missed, even at the cost of false positives.

Results:

- The model demonstrates reasonable predictive performance
- Class balancing improved detection of at-risk students

Key insight:

Low activity levels and poor performance are strong predictors of disengagement.

This model can be used to trigger alerts for academic advisors when a student shows early signs of risk.

Task 3: Recommendation System

We implemented a content-based recommendation system to suggest courses for students.

Approach:

- Courses were ranked based on average student performance
- Top-performing courses were recommended

This provides a simple but effective baseline for guiding students toward suitable academic options.

Cold-start strategy:

For new students, recommendations can be based on overall course popularity and success rates.

Future improvement:

Collaborative filtering can be used to recommend courses based on similar student profiles and behavior patterns.

Product Integration

This system can be integrated into a university dashboard used by academic advisors.

Example workflow:

- The system monitors student activity weekly
- If engagement drops significantly or predicted risk exceeds a threshold, an alert is triggered
- Advisors receive a notification with:
 - Risk score
 - Key contributing factors (low activity, missed assessments)
 - Suggested intervention actions

This allows institutions to intervene early and provide targeted support to students.

Limitations & Future Work

Limitations:

- Limited feature set due to time constraints
- Simple recommendation logic
- No advanced temporal modeling

Future improvements:

- Incorporate richer behavioral features
- Use more advanced models (e.g., gradient boosting)
- Improve recommendation system with collaborative filtering
- Add real-time monitoring and dashboard visualization